

Problem overview:

The weekly average of the natural logarithm of prices for two agricultural commodities, A and B, observed over $n=30$ weeks, follows a bivariate normal distribution given by:

$$L_i = \begin{pmatrix} Y_{i1} \\ Y_{i2} \end{pmatrix} \sim N\left(\mu \equiv \begin{pmatrix} \mu_{i1} \\ \mu_{i2} \end{pmatrix}, \Sigma \equiv \gamma^{-1}R\right) \text{ where } R = \begin{pmatrix} 1.0 & 0.5 \\ 0.5 & 1.0 \end{pmatrix}$$

$Y_{ij} \Rightarrow$ mean logprice per week for $i = 1, 2, \dots, n$ and $j = 1, 2$ for A and B

$L_i, i = 1, 2, \dots, 30 \Rightarrow$ mean logprice measurements: assumed to be independent

$\gamma \Rightarrow$ unknown common precision $\forall L_i$ measurements

Furthermore:

$E(Y_{i1}) = \mu_{i1} = \mu \Rightarrow$ unknown

$E(Y_{i2}) = \mu_{i2} = 2 \Rightarrow$ known

Since μ and γ are unknown parameters, we assume the following priors:

$$\mu|\gamma \sim N(0, 100\gamma^{-1}) \Rightarrow \frac{1}{\sqrt{2\pi 100\gamma^{-1}}} e^{-\frac{\mu^2}{2 \cdot 100\gamma^{-1}}}$$

$$\gamma \sim \text{Gamma}(\alpha, \beta) \text{ where } \alpha = \beta = 0.01 \Rightarrow \frac{0.01^{0.01}}{\Gamma(0.01)} \gamma^{0.01-1} e^{-0.01\gamma}$$

1. Explaining the suitability of the joint prior choice on (μ, γ) :

According to the conjugacy of the Normal distribution, choosing a Normal prior for μ is suitable since it ensures that the posterior for μ is Normal as well, and since it comes from the same family of distributions as the likelihood.

Since the variance of the Normal prior depends on γ^{-1} , the likelihood variance depends on γ^{-1} as well as considering the covariance matrix Σ which also consists of $\gamma^{-1}R$. Therefore, the Gamma prior is a suitable prior, since Gamma distribution is the conjugate prior of the Normal distribution.

2a. The marginal probability density function of Y_{i1} :

In order to obtain the marginal density function of Y_{i1} , we first need to find the bivariate normal density function.

The Bivariate Normal Distribution has the following form:

$$f_{L_i}(y_{i1}, y_{i2}) = \frac{1}{2\pi|\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(L_i - \mu)^T \Sigma^{-1}(L_i - \mu)} \text{ where } L_i \sim N_2(\mu, \Sigma)$$

In which the normalizing constant $2\pi|\Sigma|^{\frac{1}{2}} = 2\pi\sqrt{\sigma_1^2\sigma_2^2(1-\rho^2)}$, therefore, σ_1^2 , σ_2^2 , and ρ need to be calculated.

Since $\Sigma = \gamma^{-1} \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$, then $\text{Var}(Y_{i1}) = \sigma_{11} = \sigma_1^2 = \gamma^{-1}$ and $\text{Var}(Y_{i2}) = \sigma_{22} = \sigma_2^2 = \gamma^{-1}$

To determine the correlation coefficient ρ , we use $\rho = \frac{\sigma_{11}^2}{\sqrt{\sigma_{11}\sigma_{22}}} \Rightarrow \frac{0.5}{\sqrt{1 \cdot 1}} = 0.5$

Moreover, $\Sigma^{-1} = \frac{1}{\sigma_1^2 \sigma_2^2 (1-\rho^2)} \begin{pmatrix} \sigma_{22} & -\sigma_{12} \\ -\sigma_{21} & \sigma_{11} \end{pmatrix}$, therefore the squared statistical distance between L_i and μ can be written as:

$$\begin{aligned} (L_i - \mu)^T \Sigma^{-1} (L_i - \mu) &= \begin{pmatrix} y_{i1} - \mu \\ y_{i2} - 2 \end{pmatrix}^T \frac{1}{\sigma_1^2 \sigma_2^2 (1-\rho^2)} \begin{pmatrix} \sigma_{22} & -\sigma_{12} \\ -\sigma_{21} & \sigma_{11} \end{pmatrix} \begin{pmatrix} y_{i1} - \mu \\ y_{i2} - 2 \end{pmatrix} \\ &= \frac{1}{1-\rho^2} \left[\frac{(y_{i1}-\mu)^2}{\sigma_1^2} - 2\rho \frac{(y_{i1}-\mu)(y_{i2}-2)}{\sigma_1 \sigma_2} + \frac{(y_{i2}-2)^2}{\sigma_2^2} \right] \end{aligned}$$

Hence, the full Bivariate Normal density function is:

$$f_{Y_{i1}, Y_{i2}}(y_{i1}, y_{i2}) = \frac{1}{2\pi \sqrt{\sigma_1^2 \sigma_2^2 (1-\rho^2)}} e^{-\frac{1}{2(1-\rho^2)} \left(\frac{(y_{i1}-\mu)^2}{\sigma_1^2} - 2\rho \frac{(y_{i1}-\mu)(y_{i2}-2)}{\sigma_1 \sigma_2} + \frac{(y_{i2}-2)^2}{\sigma_2^2} \right)}$$

We can then simplify the exponent by making it a perfect square:

$$\begin{aligned} f_{Y_{i1}, Y_{i2}}(y_{i1}, y_{i2}) &= \frac{1}{2\pi \sqrt{\sigma_1^2 \sigma_2^2 (1-\rho^2)}} e^{-\frac{1}{2(1-\rho^2)} \left(\frac{(y_{i1}-\mu)^2}{\sigma_1^2} - 2\rho \frac{(y_{i1}-\mu)(y_{i2}-2)}{\sigma_1 \sigma_2} + \frac{(y_{i2}-2)^2}{\sigma_2^2} \right)} \\ &= \frac{1}{2\pi \sqrt{\sigma_1^2 \sigma_2^2 (1-\rho^2)}} e^{-\frac{1}{2(1-\rho^2)} \left[\left(\frac{y_{i1}-\mu}{\sigma_1} - \rho \frac{y_{i2}-2}{\sigma_2} \right)^2 + (1-\rho^2) \frac{(y_{i2}-2)^2}{\sigma_2^2} \right]} \\ &= \frac{1}{2\pi \sqrt{\sigma_1^2 \sigma_2^2 (1-\rho^2)}} e^{-\frac{\left(y_{i1}-\mu - \rho \frac{\sigma_1}{\sigma_2} (y_{i2}-2) \right)^2}{2\sigma_1^2 (1-\rho^2)}} e^{-\frac{(y_{i2}-2)^2}{2\sigma_2^2}} \\ &\Rightarrow \frac{1}{2\pi \sqrt{\gamma^{-1} \gamma^{-1} (1-0.5^2)}} e^{-\frac{\left(y_{i1}-\mu - 0.5 \frac{\gamma^{-1/2}}{\gamma^{-1/2}} (y_{i2}-2) \right)^2}{2\gamma^{-1} (1-0.5^2)}} e^{-\frac{(y_{i2}-2)^2}{2\gamma^{-1}}} \end{aligned}$$

Let $y_{i1}^*(y_{i2}) = \mu + \rho \frac{\sigma_1}{\sigma_2} (y_{i2} - 2)$ and similarly $y_{i2}^*(y_{i1}) = 2 + \rho \frac{\sigma_2}{\sigma_1} (y_{i1} - \mu)$

Therefore, to find the marginal PDF of Y_{i1} we need to integrate out the Y_{i2} :

$$\begin{aligned} \int_{-\infty}^{+\infty} f_{Y_{i1}, Y_{i2}}(y_{i1}, y_{i2}) dy_{i2} &= \int_{-\infty}^{+\infty} \frac{1}{2\pi \sqrt{\sigma_1^2 \sigma_2^2 (1-\rho^2)}} e^{-\frac{(y_{i2}-y_{i2}^*(y_{i1}))^2}{2\sigma_2^2 (1-\rho^2)}} e^{-\frac{(y_{i1}-\mu)^2}{2\sigma_1^2}} dy_{i2} \\ &= \frac{1}{2\pi \sqrt{\sigma_1^2 \sigma_2^2 (1-\rho^2)}} e^{-\frac{(y_{i1}-\mu)^2}{2\sigma_1^2}} \int_{-\infty}^{+\infty} e^{-\frac{(y_{i2}-y_{i2}^*(y_{i1}))^2}{2\sigma_2^2 (1-\rho^2)}} dy_{i2} \\ &= \frac{1}{2\pi \sqrt{\sigma_1^2 \sigma_2^2 (1-\rho^2)}} e^{-\frac{(y_{i1}-\mu)^2}{2\sigma_1^2}} \cdot \sqrt{2\pi} \sigma_2 \sqrt{1-\rho^2} \rightarrow \text{by the Gaussian integral} \\ &= \frac{1}{\sqrt{2\pi} \sigma_1} e^{-\frac{(y_{i1}-\mu)^2}{2\sigma_1^2}} \Rightarrow Y_{i1} \sim N(\mu, \sigma_1^2) \end{aligned}$$

Hence, since $\sigma_1^2 = \gamma^{-1}$

$$\sqrt{\frac{\gamma}{2\pi}} e^{-\frac{\gamma}{2}(y_{i1}-\mu)^2} \Rightarrow Y_{i1} \sim N(\mu, \gamma^{-1})$$

The marginal pdf for Y_{i2} can be obtained using the same technique: $f_{Y_{i2}}(y_{i2}) \sim N(2, \gamma^{-1})$

2b. According to the joint PDF $f_{Y_{i1}, Y_{i2}}(y_{i1}, y_{i2})$ from 2a, based on the factorization of the joint PDF of a Bivariate Normal, we obtain the following:

$$f_{Y_{i1}, Y_{i2}}(y_{i1}, y_{i2}) = \frac{1}{\sqrt{2\pi\sigma_1^2(1-\rho^2)}} e^{-\frac{(y_{i1}-y_{i1}^*(y_{i2}))^2}{2\sigma_1^2(1-\rho^2)}} \times \frac{1}{\sqrt{2\pi\sigma_2^2}} e^{-\frac{(y_{i2}-2)^2}{2\sigma_2^2}}$$

$$\Rightarrow f_{Y_{i1}|Y_{i2}}(y_{i1}|y_{i2}) = \frac{f_{Y_{i1}, Y_{i2}}(y_{i1}, y_{i2})}{f_{Y_{i2}}(y_{i2})} = \frac{1}{\sqrt{2\pi\sigma_1^2(1-\rho^2)}} e^{-\frac{(y_{i1}-y_{i1}^*(y_{i2}))^2}{2\sigma_1^2(1-\rho^2)}} = \sqrt{\frac{\gamma}{2\pi(1-0.5^2)}} e^{-\frac{\gamma}{2 \cdot 0.5^2} \left(y_{i1} - \left(\mu + 0.5 \frac{\gamma^{-1}}{\gamma^{-1}} (y_{i2}-2) \right) \right)^2}$$

Therefore, the conditional distribution of $Y_{i1}|Y_{i2} = y_{i2}$ is:

$$N(\mu_{Y_{i1}|Y_{i2}} = \mu + 0.5(y_{i2} - 2), \sigma_{Y_{i1}|Y_{i2}}^2 = \gamma^{-1}(1 - 0.5^2))$$

3a. The likelihood corresponding to the unknown parameters μ and γ given the observed data L :

As derived in question 2, the full joint pdf of the Bivariate Normal is given as:

$$f(L_i | \mu, \gamma) = \frac{1}{2\pi|\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(L_i - \mu)^T \Sigma^{-1} (L_i - \mu)} = \frac{1}{2\pi(\gamma^{-2} 0.75)^{1/2}} e^{-\frac{1}{2} \begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix}^T \frac{\gamma}{0.75} \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix} \begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix}}$$

Where:

$$\Sigma^{-1} = \frac{\gamma}{0.75} \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix} \Rightarrow by (kA)^{-1} = \frac{1}{k} A^{-1}$$

$$|\Sigma| = \gamma^{-2} |R| \rightarrow by \det(2A) = 2^n |A| \Rightarrow |R| = 0.75 \Rightarrow |\Sigma| = \gamma^{-2} 0.75$$

Therefore, the likelihood is as follows:

$$L(L_i | \mu, \gamma) = \prod_{i=1}^n f(L_i | \mu, \gamma) = \left(\frac{1}{2\pi(\gamma^{-2} 0.75)^{1/2}} \right)^n e^{-\frac{\gamma}{2 \cdot 0.75} \sum_{i=1}^n \left[\begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix}^T \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix} \begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix} \right]}$$

3b. The full posterior distribution of (μ, γ) :

The posterior distribution function has the following form:

$$\pi(\mu, \gamma | L_i) \propto L(L_i | \mu, \gamma) \cdot \pi(\mu | \gamma) \cdot \pi(\gamma)$$

$$\text{Where: } \pi(\mu | \gamma) = \frac{1}{\sqrt{2\pi 100\gamma^{-1}}} e^{-\frac{\mu^2}{2 \cdot 100\gamma^{-1}}} \Rightarrow \text{prior on } \mu | \gamma \text{ and } \pi(\gamma) = \frac{0.01^{0.01}}{\Gamma(0.01)} \gamma^{0.01-1} e^{0.01\gamma} \Rightarrow \text{prior on } \gamma$$

$$\therefore \pi(\mu, \gamma | L_i) \propto \left(\frac{1}{2\pi(\gamma^{-2} 0.75)^{1/2}} \right)^n e^{-\frac{\gamma}{2 \cdot 0.75} \sum_{i=1}^n \left[\begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix}^T \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix} \begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix} \right]} \cdot \frac{1}{\sqrt{2\pi 100\gamma^{-1}}} e^{-\frac{\mu^2}{2 \cdot 100\gamma^{-1}}} \cdot \frac{0.01^{0.01}}{\Gamma(0.01)} \gamma^{0.01-1} e^{0.01\gamma}$$

$$\propto \gamma^n e^{-\frac{2\gamma}{3} \sum_{i=1}^n \left[\begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix}^T \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix} \begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix} \right]} \cdot \gamma^{\frac{1}{2}} e^{-\frac{\mu^2 \gamma}{200}} \cdot \gamma^{0.01-1} e^{0.01\gamma}$$

$$\propto \gamma^{n+\frac{1}{2}+0.01-1} e^{-\gamma \left[\frac{2}{3} \sum_{i=1}^n \left[\begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix}^T \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix} \begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix} \right] + \frac{\mu^2}{200} + 0.01 \right]}$$

4. Identifying and deriving the conditional posterior distribution of $(\mu|\gamma, L)$:

Using the hint: Based on question 2, since Y_{i1} and Y_{i2} are both observed, the likelihood for μ is determined by the conditional distribution $Y_{i1}|Y_{i2}$. Therefore, the conditional posterior probability density function of $(\mu|\gamma, L)$ depends on the conditional PDF derived in question 2.

Therefore, we can derive the conditional posterior of $(\mu|\gamma, L)$ as follows:

Since $Y_{i1}|Y_{i2} \sim N(\mu + 0.5(y_{i2} - 2), 0.75\gamma^{-1})$ where $\pi(\mu|\gamma) = \frac{1}{\sqrt{2\pi 100\gamma^{-1}}} e^{-\frac{\mu^2}{2 \cdot 100\gamma^{-1}}}$, we can find the posterior distribution function:

$$L(\mu, \gamma|L) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi 0.75\gamma^{-1}}} e^{-\frac{(y_{i1} - (\mu + 0.5(y_{i2} - 2)))^2}{2 \cdot 0.75\gamma^{-1}}} = \left(\frac{1}{\sqrt{2\pi 0.75\gamma^{-1}}} \right)^n e^{-\frac{\sum_{i=1}^n (y_{i1} - (\mu + 0.5(y_{i2} - 2)))^2}{2 \cdot 0.75\gamma^{-1}}}$$

$$\therefore \pi(\mu|\gamma, L) \propto L(\mu, \gamma|L) \cdot \pi(\mu|\gamma)$$

$$\begin{aligned} &\propto \frac{\gamma^{\frac{n}{2}}}{(2\pi 0.75)^{\frac{n}{2}}} e^{-\frac{\gamma}{2 \cdot 0.75} \sum_{i=1}^n (y_{i1} - (\mu + 0.5(y_{i2} - 2)))^2} \cdot \frac{\gamma}{\sqrt{200\pi}} e^{-\frac{\mu^2 \gamma}{200}} \\ &\propto e^{-\gamma \left[\frac{1}{1.5} \sum_{i=1}^n (y_{i1} - (\mu + 0.5(y_{i2} - 2)))^2 + \frac{\mu^2}{200} \right]} \end{aligned}$$

Similarly, the conditional posterior could be found through isolating the terms independent of μ from the full posterior distribution:

$$\begin{aligned} \pi(\mu, \gamma|L_i) &\propto \left(\frac{1}{2\pi(\gamma^{-2} 0.75)^{\frac{1}{2}}} \right)^n e^{-\frac{\gamma}{2 \cdot 0.75} \sum_{i=1}^n \left[\begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix}^T \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix} \begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix} \right]} \cdot \frac{1}{\sqrt{2\pi 100\gamma^{-1}}} e^{-\frac{\mu^2}{2 \cdot 100\gamma^{-1}}} \cdot \frac{0.01^{0.01}}{\Gamma(0.01)} \gamma^{0.01-1} e^{0.01\gamma} \\ \Rightarrow \pi(\mu|\gamma, L) &\propto e^{-\gamma \left[\frac{1}{1.5} \sum_{i=1}^n \left[\begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix}^T \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix} \begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix} \right] + \frac{\mu^2}{200} \right]} \end{aligned}$$

However, in order to be able to use the parameters of this conditional posterior, the distribution needs to be rearranged:

$$\begin{aligned} \pi(\mu|\gamma, L) &\propto e^{-\gamma \left[\frac{1}{1.5} \sum_{i=1}^n (y_{i1} - (\mu + 0.5(y_{i2} - 2)))^2 + \frac{\mu^2}{200} \right]} \\ &\propto e^{-\frac{\gamma}{2} \left[\frac{1}{0.75} \sum_{i=1}^n (y_{i1} - (\mu + 0.5(y_{i2} - 2)))^2 + \frac{\mu^2}{100} \right]} \end{aligned}$$

Let $X_i = (y_{i1} + 0.5(y_{i2} - 2))$, therefore:

$$\begin{aligned} &\propto e^{-\frac{\gamma}{2} \left[\frac{1}{0.75} \sum_{i=1}^n (X_i - \mu)^2 + \frac{\mu^2}{100} \right]} \\ &\propto e^{-\frac{\gamma}{2} \left[\frac{\sum_{i=1}^n X_i}{0.75} - \frac{2 \sum_{i=1}^n X_i \mu}{0.75} + \frac{n\mu^2}{0.75} + \frac{\mu^2}{100} \right]} \\ &\propto e^{-\frac{\gamma}{2} \left[\mu^2 \left(\frac{n}{0.75} + \frac{1}{100} \right) - \frac{2 \sum_{i=1}^n X_i \mu}{0.75} \right]} \end{aligned}$$

Let $a = \left(\frac{n}{0.75} + \frac{1}{100} \right)$ and $b = \frac{\sum_{i=1}^n X_i}{0.75}$ and therefore:

$$\begin{aligned} &\propto e^{-\frac{\gamma}{2} [\mu^2 a - 2b\mu]} \\ &\propto e^{-\frac{\gamma a}{2} [\mu^2 - 2\frac{b}{a}\mu]} \\ &\propto e^{-\frac{\gamma a}{2} [\mu^2 - 2\frac{b}{a}\mu + \left(\frac{b}{a}\right)^2 - \left(\frac{b}{a}\right)^2]} \\ &\propto e^{-\frac{\gamma a}{2} [\mu^2 - 2\frac{b}{a}\mu + \left(\frac{b}{a}\right)^2]} \\ &\propto e^{-\frac{\gamma a}{2} \left(\mu - \frac{b}{a} \right)^2} \end{aligned}$$

Hence after substituting all of the values previously obtained we get the following:

$$\begin{aligned}\pi(\mu|\gamma, L) &\propto e^{-\frac{\gamma}{2}\left(\frac{n}{0.75} + \frac{1}{100}\right)\left(\mu - \frac{\frac{\sum_{i=1}^n X_i}{0.75}}{\frac{n}{0.75} + \frac{1}{100}}\right)^2} \\ &\propto e^{-\frac{\gamma}{2}\left(\frac{30}{0.75} + \frac{1}{100}\right)\left(\mu - \frac{\frac{\sum_{i=1}^n (y_{i1} + 0.5(y_{i2} - 2))}{0.75}}{\frac{30}{0.75} + \frac{1}{100}}\right)^2} \sim N\left(\frac{\sum_{i=1}^{30} (y_{i1} + 0.5(y_{i2} - 2))}{30.0075}, \frac{1}{\gamma(40.01)}\right)\end{aligned}$$

Hence, we obtain the conditional posterior distribution of $(\mu|\gamma, L)$.

5. In order to obtain the conditional posterior distribution of $(\gamma|\mu, L)$, we can derive it from the full posterior distribution $\pi(\mu, \gamma|L_i)$ – which was done in 3b – by eliminating everything independent of γ .

Therefore, we get the following:

$$\begin{aligned}\pi(\gamma|\mu, L_i) &\propto \left(\frac{1}{2\pi(\gamma^{-2}0.75)^{\frac{1}{2}}}\right)^n e^{-\frac{\gamma}{2 \cdot 0.75} \sum_{i=1}^n \left[\begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix}^T \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix} \begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix} \right]} \cdot \frac{1}{\sqrt{2\pi 100\gamma^{-1}}} e^{-\frac{\mu^2}{2 \cdot 100\gamma^{-1}}} \cdot \frac{0.01^{0.01}}{\Gamma(0.01)} \gamma^{0.01-1} e^{0.01\gamma} \\ \Rightarrow \pi(\gamma|\mu, L) &\propto \gamma^{n+0.51-1} e^{-\gamma \left[\frac{2}{3} \sum_{i=1}^n \left[\begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix}^T \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix} \begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix} \right] + \frac{\mu^2}{200} + 0.01 \right]} \\ &\sim \text{Gamma}\left(30 + 0.51, \frac{2}{3} \sum_{i=1}^{30} \left[\begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix}^T \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix} \begin{pmatrix} Y_{i1} - \mu \\ Y_{i2} - 2 \end{pmatrix} \right] + \frac{\mu^2}{200} + 0.01\right)\end{aligned}$$

6. Description of the Gibbs sampling algorithm:

a. Initialization:

1. generate the initial value for $\gamma^{(0)}$ from its prior distribution $\gamma \sim \text{Gamma}(0.01, 0.01)$
2. generate the initial value for $\mu^{(0)}$ from its prior distribution $\mu|\gamma \sim N(0, 100\gamma^{-1(0)})$

b. Iterate N times, for $n = 1, 2, \dots, N$

1. update $\gamma^{(n)}$ using:

$$\pi(\gamma^{(n)}|\mu^{(n-1)}, L_i) \sim \text{Gamma}\left(30 + 0.51, \frac{2}{3} \sum_{i=1}^{30} \left[\begin{pmatrix} Y_{i1} - \mu^{(n-1)} \\ Y_{i2} - 2 \end{pmatrix}^T \begin{pmatrix} 1 & -0.5 \\ -0.5 & 1 \end{pmatrix} \begin{pmatrix} Y_{i1} - \mu^{(n-1)} \\ Y_{i2} - 2 \end{pmatrix} \right] + \frac{\mu^{2(n-1)}}{200} + 0.01\right)$$

2. update $\mu^{(n)}$ using:

$$\pi(\mu^{(n)}|\gamma^{(n)}, L) \sim N\left(\frac{\sum_{i=1}^{30} (y_{i1} + 0.5(y_{i2} - 2))}{30.0075}, \frac{1}{\gamma^{(n)}(40.01)}\right)$$

c. The samples can be taken from a Gibbs algorithm after the burn-in period, since the first n samples are believed to be biased since they are highly dependent on the initial values. After n iterations, the chain starts to converge to stationarity and is therefore no longer biased.

7. Based on the MATLAB code, the numerical summary consisting of posterior means, standard deviations and 95% credible intervals of the components of μ and γ are the following:

Numerical Summaries:				
Parameter	Posterior Mean	Posterior SD	CI – 2.5%	CI – 97.5%
μ	3.3929	0.29328	2.812	3.972
γ	0.30203	0.05572	0.20326	0.42027

Graphical evidence that the Gibbs chain has mixed satisfactorily:

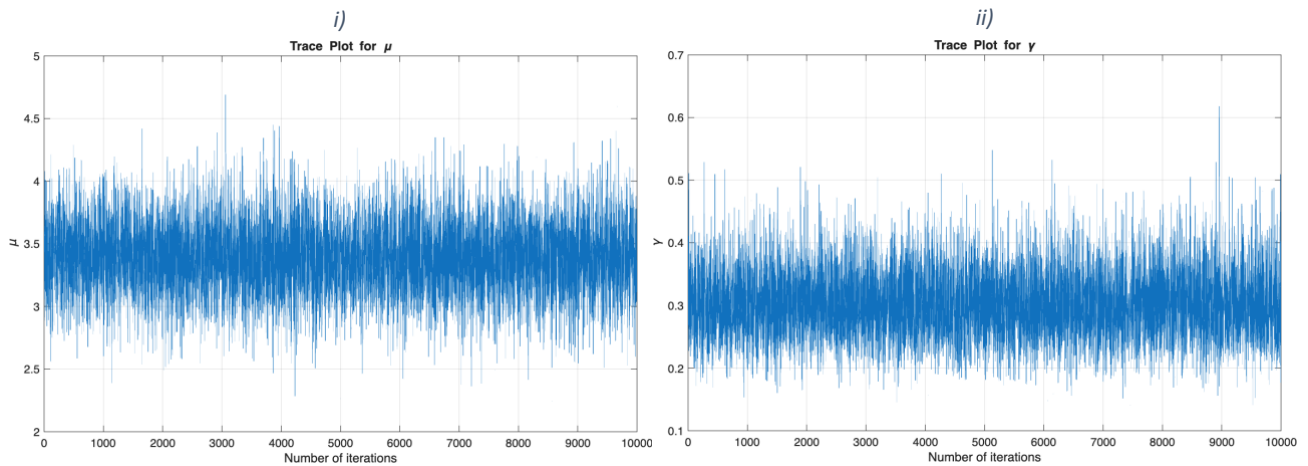


Figure 1: (i) Trace Plot for μ ; (ii) Trace Plot for γ

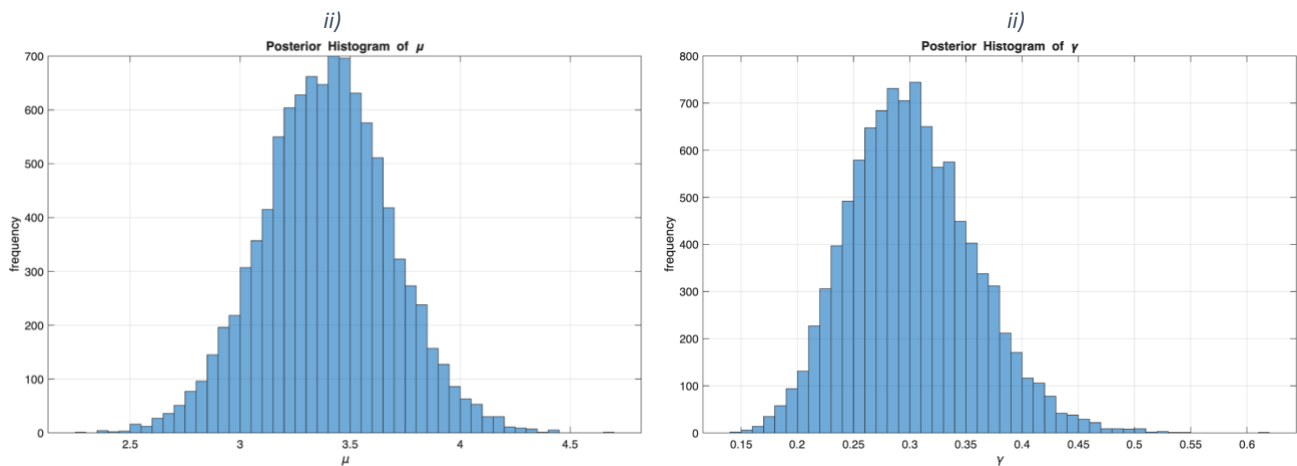


Figure 2: (i) Posterior Histogram for μ ; (ii) Posterior Histogram for γ

Comment on μ : Based on the previous parts of the report, μ follows a Normal distribution. According to Figure 2(i), the histogram of μ appears to be normally distributed and is centered around 3.4. Based on Figure 1(i), the chains fluctuate around the mean of 3.4. The trace plot seems to have a higher variability compared to the traceplot of γ in Figure 1(ii).

Comment on γ : It was observed that γ follows a Gamma distribution. According to Figure 2(ii), the histogram of γ appears to be slightly skewed to the right, and based on Figure 1(ii), the chains fluctuate around the mean of 0.3. The trace plot appears to have less variability compared to Figure 1(i).

Both of the traceplots appear to be stable and stationary with no visible trends, therefore indicating that the Gibbs sampler has converged. The posterior histograms displayed in Figure 2 indicate a stable distribution. Therefore, the Gibbs chain has mixed satisfactorily.